INSULIN LISPRO

High-alert drug

U-100 and U-200 — check

CHEMICAL NAME B28-lysine, B29-proline human insulin (recombinant insulin analogue)
CLASS Rapid-acting insulin analogue
TRADE NAMES Humalog · Admelog · Lyumjev (ultra-rapid) · Liprolog

THERAPEUTIC USES / INDICATIONS

- Prandial insulin in type 1 and type 2 diabetes.
- Correction doses.
- Insulin pump therapy.
- Gestational diabetes (with basal insulin).

PHARMACODYNAMICS

- Insulin receptor agonist; rapid, short action mimicking physiological meal response.

ADMINISTRATION

- **SC** 0–15 min before meals (or immediately after in children).
- U-200 pens: only use with the dedicated pen — never draw up into a syringe.
- IV use in hospital where approved.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Lipohypertrophy, injection-site reactions.
- Weight gain.

ADVERSE EFFECTS (SERIOUS)

- Hypoglycaemia; hypokalaemia.
- Dosing errors with concentrated U-200 product.

DOSAGES

- **Individualised** (as insulin aspart): mealtime doses by carbohydrate ratio and correction factor.
- Type 2 (adding mealtime insulin): start at one main meal with a small dose and titrate to glucose (local protocol).

PHARMACOKINETICS

- **Onset:** 15 min. **Peak:** 0.5–1.5 h. **Duration:** 3–5 h.
- Reversal of B28/B29 residues prevents self-association → rapid absorption.

CONTRAINDICATIONS & CAUTIONS

- Hypoglycaemia; hypersensitivity.

KEY INTERACTIONS

- As other insulins.

NURSING MANAGEMENT

- Confirm concentration and device.
- Give with meal; monitor pre-meal glucose.
- Assess injection sites for lipohypertrophy.

PATIENT EDUCATION

- Inject at mealtimes; never skip food after a dose.
- Rotate injection sites.
- Check glucose regularly; carry glucose.

KEY POINTS

- Rapid-acting analogue; onset 15 min.
- U-200: pen only, never syringe.
- Mealtime insulin.
- Hypoglycaemia without food.

SOURCES CHECKED **[594]** Humalog 100 units/ml Junior KwikPen, solution for injection in a pre-filled pen (UK SmPC)

Full references p. 432–457